3D modelling of confined gastruloid aggregates

November 3, 2026

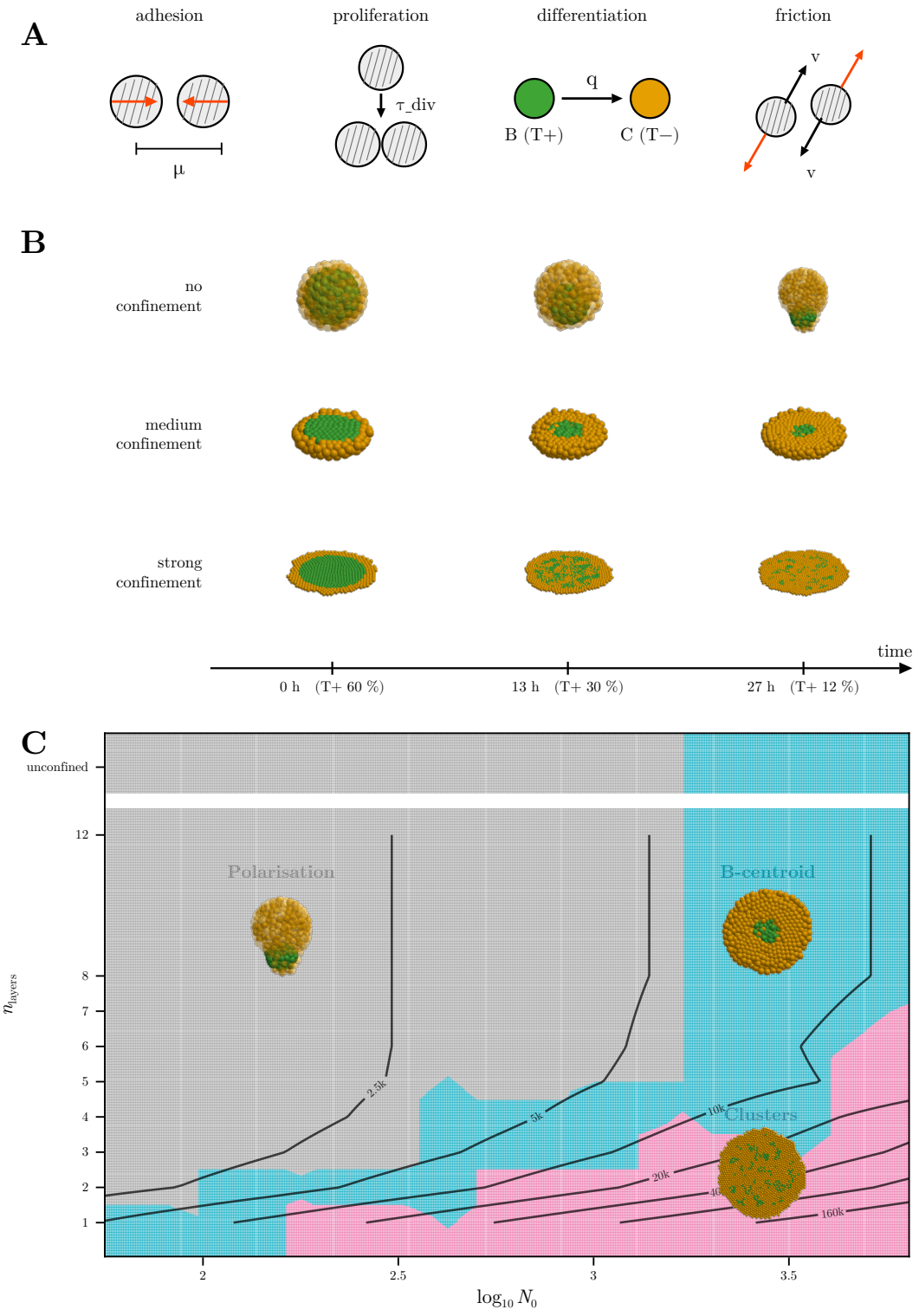


Figure 1: Overview. (A) Model ingredients. (B) Evolution of same aggregate ($N_0 = 600$ cells) without confinement, with medium confinement (3 layers) and with strong confinement (1 layer). (C) Outcome phase space: number of initial cells against the number of layers between the plates, read when 12% of the cells are B. Lines: area of the pancake at squeezing (μm^2). Each example sits on its region: the last frames of the rows of (B).

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To-Do

1. Check all the figures.
2. Make a comparison table with diameter and number of cells accoutning for some overlapping
3. Add how we comute length in the model
4. Larger aggregates ($N_0 = 1200$ to 4800) still have a central cluster when 20% of the cells are B, and the largest never polarise.

1 The experiment

In the expriments, they grow a gastruloid and confine it right before polarisation should occur. Cells transition $A \to B \to C$, and in that moment the distribution is something like: almost no A-cells, a centroid of B-cells, and an outer layer mainly formed of C-cells. On squeezing the aggregate between two plates, there are three possible outcomes.

1. If the confinement is strong enough (i.e. small plate distance), the centroid starts disintegrating usually forming a ring-like structure and then breaks into clusters.
2. For medium confinement, the centroid does not polarise nor breaks into clusters and eventually differentiates away. It may form a ring-like pattern but without breaking.
3. For weak confinement, the centroid polarises as when there is no confinement.

Determining whether confinement is strong or weak is not trivial and is actually key to this research. The experiments observe that the larger the intitial size on squeezing, the most layers we also get clusters with.

The experimentalists image the aggregate and track the cells in order to perform certain measurements.

Spatial cluster quantification. It is used to determine whether clusters have formed.

Spatial clustering was quantified using a density-based clustering approach adapted from DBSCAN (Density-Based Spatial Clustering of Applications with Noise). Clusters were identified based on the local enrichment and spatial connectivity of cells belonging to the cell type B population. For each cell, the fraction of neighbouring cells belonging to the same population was calculated within a defined three-dimensional Euclidean radius of 13 µm, considering all surrounding cells as the reference. Cells for which at least 40% of neighbouring cells belonged to the population of interest were defined as core cells. Core cells were grouped into the same cluster if they were directly or indirectly connected through other core cells within a maximum pairwise distance of 13 µm. Cells belonging to the population of interest that did not fulfil the core-cell criterion were assigned as border cells to the cluster of their nearest core cell, provided that this core cell was located within 13 µm. Only clusters containing at least 20 cells, including

core and border cells, were retained for further analysis. Clustering was quantified based on the number of clusters, the fraction of cells assigned to clusters, cluster sizes, and the effective number of clusters (N_{eff}), which accounts for differences in cluster size distribution. The latter was calculated as $N_{\text{eff}} = \frac{1}{\sum p_i^2}$, where p_i represents the fraction of all clustered cells belonging to cluster i, providing a measure of cluster number that accounts for differences in cluster size.

Polarisation. Polarisation is defined as follows. Let r be the center of mass of the aggregate, namely $r = \frac{1}{N}\sum r_i$. Let r_c be the center of mass of the centroid. Let R be the half of the length of the PC1 axis (longest axis). Then polarisation P is given by

$$P = \frac{|r - r_c|}{R}$$

The cutoff is set to 0.3. To determine it we just looked at the polarisation histogram of control gastruloids. Here is a plot with samples of different sizes. Where you a see that the polarisation degree at that timepoint seems to be also size dependent, but 0.3 includes the majority.

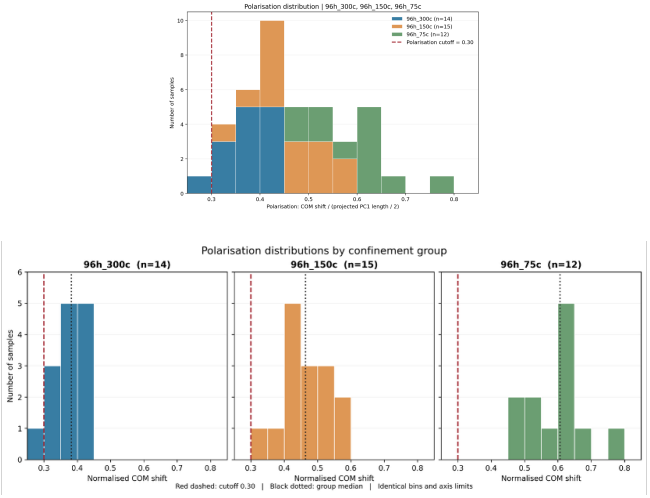


Figure 2: Polarisation histogram of control gastruloids.

Figures

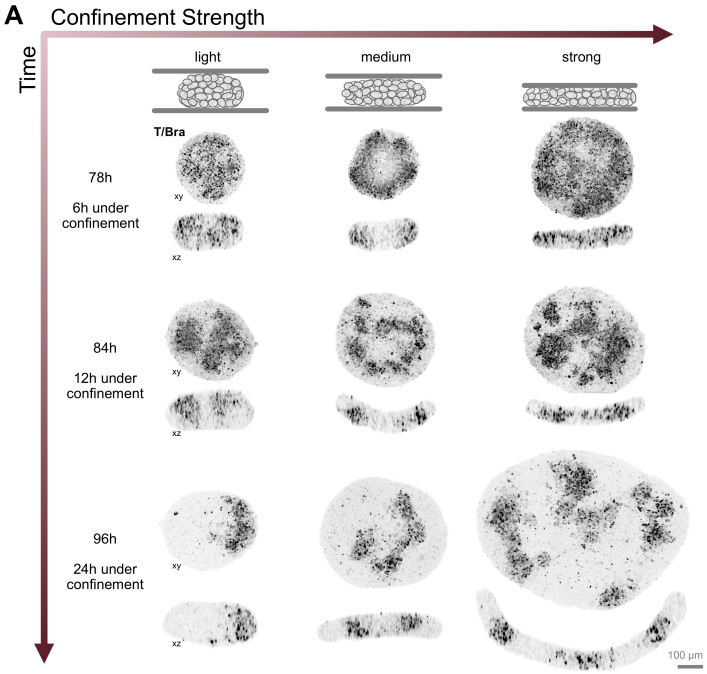


Figure 3: T/Bra patterning under light, medium and strong confinement (columns) at 78 h, 84 h and 96 h, i.e. 6 h, 12 h and 24 h under confinement (rows). Each aggregate is shown from the top (xy) and from the side (xz). Light confinement keeps a pole, medium confinement keeps the T/Bra cells in the centre and strong confinement breaks them into clusters. From V. Weichselberger's poster.

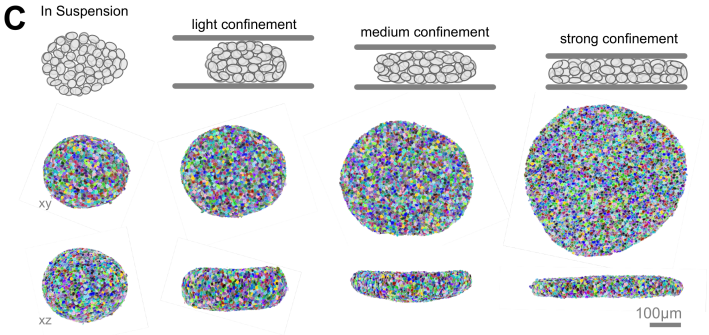


Figure 4: 3D segmentation of gastruloids after 24 h in suspension or under light, medium and strong confinement, seen from the top (xy) and from the side (xz).

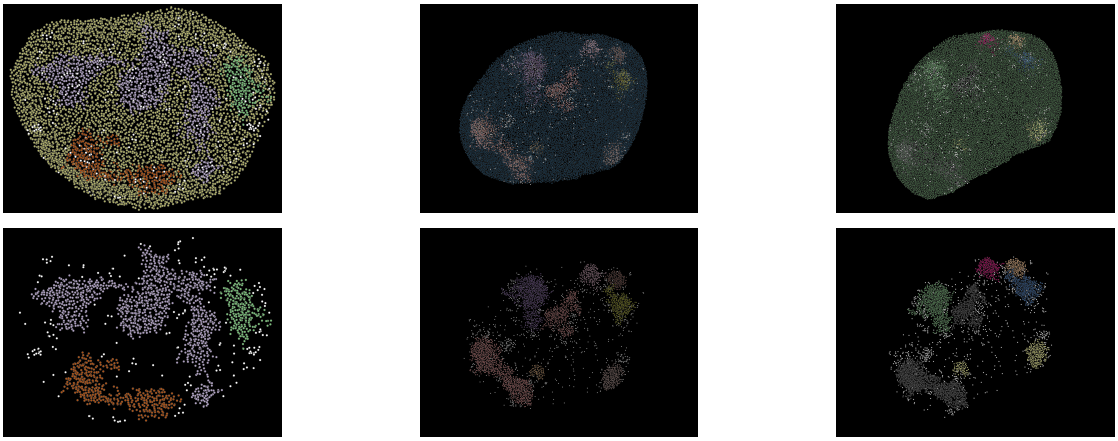


Figure 5: Spatial cluster quantification on three experimental aggregates. Top: all cells, coloured by the cluster they are assigned to. Bottom: only the cells assigned to clusters.

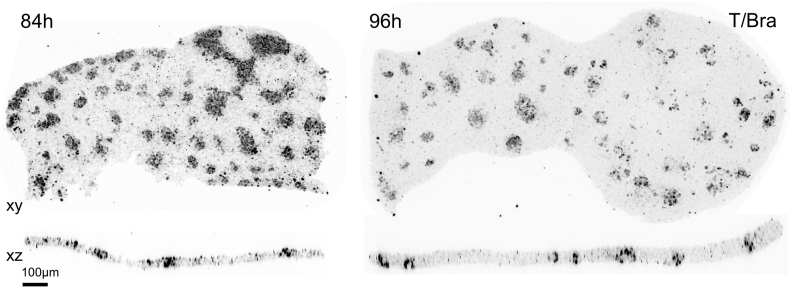


Figure 6: T/Bra clusters in gastruloids under extreme confinement at 84 h and 96 h (preliminary data), seen from the top (xy) and from the side (xz). From V. Weichselberger's poster.

2 The physical description

2.1 Simplificating reality

First, we grow and differentiate a gastruloid aggregate before polarisation begins. Instead of the three-state description used in Oriola et al. 2026 ($A \to B \to C$), we assume all A cells have differentiated away at the moment of squeezing and therefore only consider B (T^+) and C (T^-) cells.

We define out initial aggregate to be a spheroid with a central cluster of B cells containing about 60% of the cells in the aggregate. The rest of the cells are in state C. If we were to let this further evolve, this cluster would polarise.

In order to model the behaviour of the confined aggregate, we won't study the confinement process but instead consider an already-squeezed aggregate with a B-centroid. We define it by growing an aggregate under confinement, setting the inital cells as they would result on confining the spherical gastruloid, letting the cells relax, and then starting differentiaion and proliferaiton.

We want to replicate all three behaviours described previously.

Fusion. We use fusion as a sanity check to see that they are not only replicating the confined patterning, just like we do with unconfined polarisation. We see that, under certain conditions, two modelled gasruloids fuse as they do experimentally. In particular, we see that

1. Two spherical B-only aggregates fuse perfectly forming a single sphere.
2. A B-aggregate and a C-aggregate glue together but do not form a single sphere.

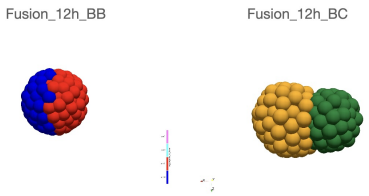


Figure 7: Expected fusion behaviour.

2.2 Modelling 3D aggregates

Mechanics. In the agent-based simulations, a single cell i is represented by the position of its centre of mass $\mathbf{x}_i$ and its dynamics is given by the following overdamped equations of motion:

$$\begin{aligned} \dot{\mathbf{x}}_i &= \mathbf{v}_i \,, \\ \mathbf{v}_i &= \frac{1}{\lambda n_i} \sum_j \mathbf{F}_{ij} + \frac{\alpha}{n_i} \sum_j \mathbf{v}_j \,. \end{aligned}$$

where isotropic friction forces are assumed and $\mathbf{v}_i$ is the cell velocity. λ_s is the friction coefficient for the cell-substrate or cell-ECM interaction, while λ_r is the relative cell-cell friction coefficient.

The passive force $\mathbf{F}_{ij}$ describes cell-cell adhesion and repulsion that depends on the position of the two interacting cells (i, j) as well as their states $(s_i, s_j) \in \{B, C\}^2$:

$$\mathbf{F}_{ij}(s_i, s_j, d_{ij}) = \begin{cases} f(d_{ij})\, \hat{\mathbf{r}}_{ij} F_{\text{rep}}^{(s_i,s_j)}, & \text{if } d_{ij} \le 2r \\ f(d_{ij})\, \hat{\mathbf{r}}_{ij} F_{\text{att}}^{(s_i,s_j)}, & \text{if } 2r < d_{ij} < 2\mu r \\ 0, & \text{otherwise} \,, \end{cases}$$

where $d_{ij} = |\mathbf{x}_i - \mathbf{x}_j|$, $\hat{\mathbf{r}}_{ij} = \frac{\mathbf{x}_i - \mathbf{x}_j}{d_{ij}}$, forces $F_{\text{rep}}^{(s_i,s_j)}$, $F_{\text{adh}}^{(s_i,s_j)}$ are 3×3 matrices that describe the magnitude of the interaction, and

$$f(d_{ij}) = \left(\frac{2r}{d_{ij}} - 1\right)\left(\frac{2\mu r}{d_{ij}} - 1\right).$$

Remark 1. *In the code we set $\hat{\alpha} = \lambda\alpha$ and so*

$$\mathbf{v}_i = \frac{1}{\lambda n_i}\left(\sum_j \mathbf{F}_{ij} + \alpha \sum_j \mathbf{v}_j\right).$$

Division. Cells undergo division according to the following rules:

- A division axis is randomly chosen over the unit sphere.
- The two daughter cells are generated with equal radius as the parent cell and positioned on opposite sides of the division axis which is centred at the parent cell's location. The daughter cells are generated with their centre of mass at a fixed distance from that of the parent cell.
- A new division time is then assigned to each daughter cell, sampled from

$$\mathcal{U}\big(\tau_{\text{div}}(1 - \sigma_{\text{div}}),\ \tau_{\text{div}}(1 + \sigma_{\text{div}})\big),$$

Differentiaton. The transition rate from B to C is constant,

$$r_{B\to C} \equiv q\,,$$

so every B cell becomes C with probability $q\,\Delta t$ per time step.

Confinement. The plates sit at $z = 0$ and $z = L$, with $L = 2.1\,r\times$ (number of layers). They are rigid: a cell touching a plate cannot move further into it, so the slab keeps its thickness during the whole simulation. A cell closer than 0.05 to a plate also feels plate friction, an extra in-plane drag:

$$\big(\lambda n_i\,\mathbb{I} + \lambda_{\text{wall}}\,\mathbb{P}\big)\,\mathbf{v}_i = \sum_j \mathbf{F}_{ij} + \alpha \sum_j \mathbf{v}_j\,, \qquad \mathbb{P} = \mathbb{I} - \hat{\mathbf{z}}\hat{\mathbf{z}}^{\top}\,,$$

where $\mathbb{P}$ projects onto the xy-plane, so the extra drag acts only parallel to the plates.

This plate friction stands for adhesion to, or rubbing on, the confining surfaces.

2.3 Analysis of the model

Remark 2. *Pulling two attached cells apart costs the* ***adhesion energy***

$$W_{XY} = \int_{2r}^{2\mu r} |F|\,\mathrm{d}d = F_{att}^{(X,Y)}\,2r\,A(\mu_{XY})\,, \qquad A(\mu) = (1+\mu)\ln\mu - 2(\mu - 1)\,.$$

These energies set the surface tensions $\sigma_X = W_{XX}/2$ and the interfacial tension $\gamma_{BC} = (W_{BB} + W_{CC})/2 - W_{BC}$. Two aggregates glue without rounding into one sphere when $|\sigma_B - \sigma_C| < \gamma_{BC} < \sigma_B + \sigma_C$.

3 The simulations

The main challenge of the simulations is to replicate all the aforementioned behaviours using the same model. All the parameters are in Table 2.



Figure 8: Initial condition.

Initial condition. We initialise a B-centroid containing 60% of the cells in the aggregate; the rest are set tp C. If the aggregate is confined, we let the system relax after changing the state of the cells to account for the differential adhesion between cell states. See Figure 8.

Fusion. We place two aggregates close in space and let the mechanics evolve without division nor differentiation (see Figure 9). We observe that the BB and CC cases fuse, whereas BC glues, as expected. This is a sanity check to see the gastruloids are replicating behaviours we know and not used in the experiments.

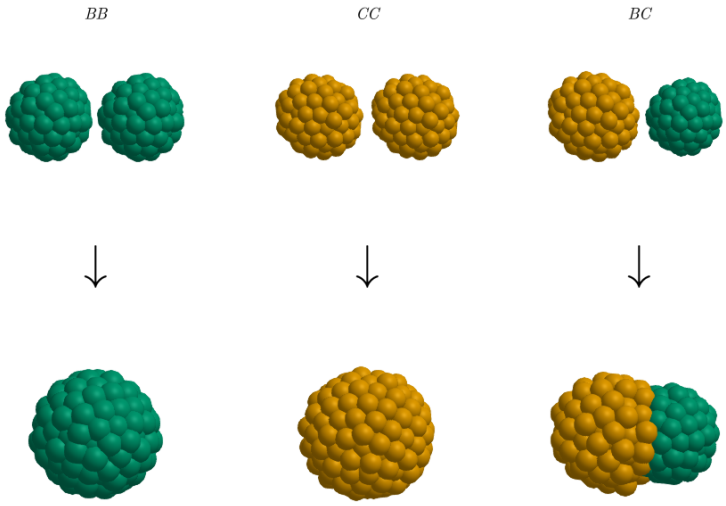


Figure 9: Fusion assay after 24 hours: BB, CC and BC doublets.

Polarisation of unconfined aggreagtes Unconfined aggregates of up to $N_0 = 600$ cells polarise before B runs out (Figure 10). Larger aggregates ($N_0 = 1200$ to 4800) still have a central cluster when 20% of the cells are B, and the largest never polarise.

Confinement Given an initial number of cells, we study the evolution of the aggregate under strong, medium and weak confinement (see Figure 11 for $N_0 = 600$).

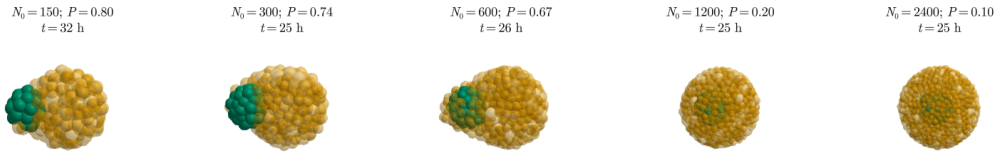


Figure 10: Polarisation..

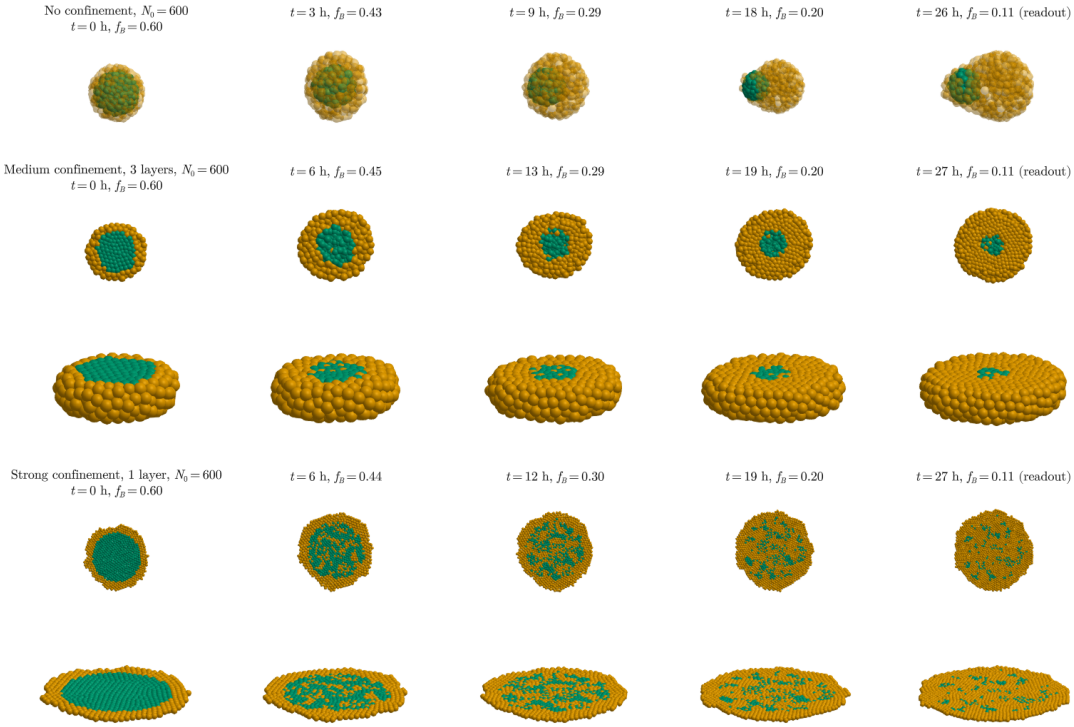


Figure 11: Evolution of a 600-cell aggregate for weak, medium and strong confinement.

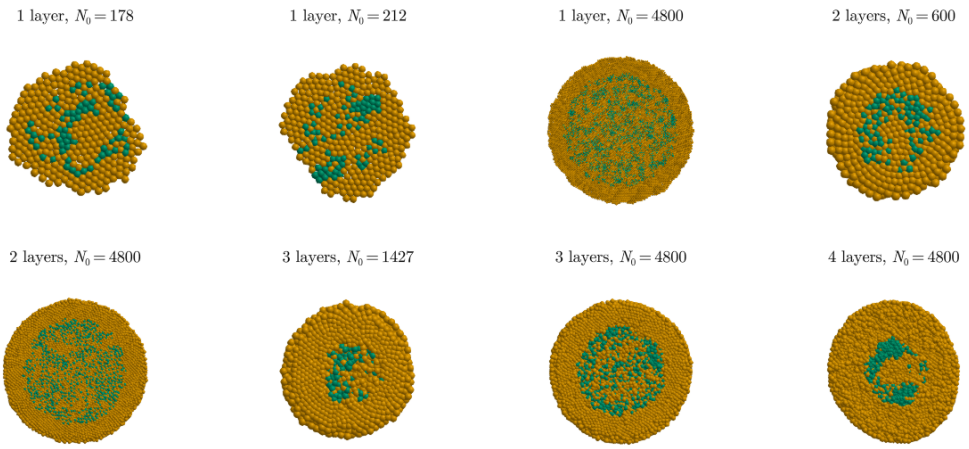


Figure 12: Example patterns for different initial conditions.

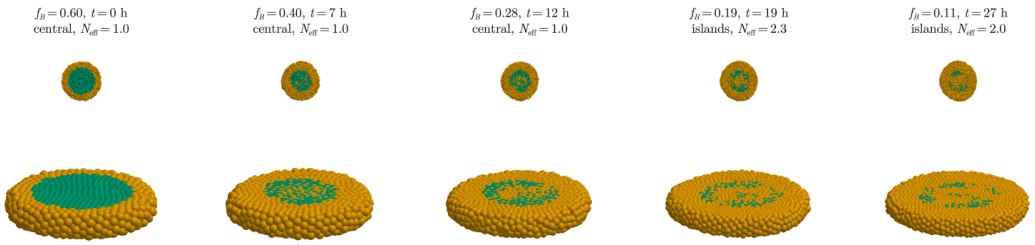


Figure 13: Three layers, $N_0 = 2400$. Central cluster forming a ring an then breaking and forming patterns.

3.1 Phase space

Finally, we compute the phase space relating the outcome, the number of layers and the number of cells N_0 at the moment of squeezing (Figure 14), with 2 or 3 simulations per point (178 in total). The outcome is read when 20% of the cells are B (96 h in the experiments) and classified as central cluster, pattern (a ring or islands), polarised, or other.

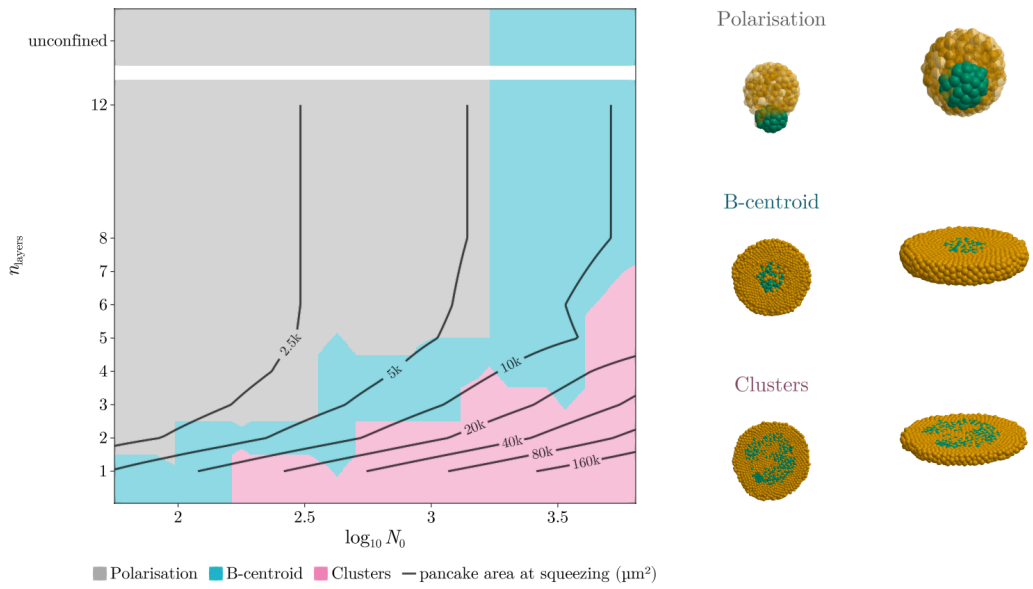


Figure 14: Phase space, read when 12% of the cells are B (about 26 h after the start of differentiation; Section 3.2). The colour is the majority outcome of the nearest simulated point (a tie between seeds goes to the outcome at the end of the simulations, 10%). Lines: area of the pancake at squeezing (μm^2). Right: one example of each outcome (8 layers, $N_0 = 300$; 3 layers, $N_0 = 1200$; 2 layers, $N_0 = 1200$).

The threshold for patterns grows quickly with the thickness, from $N_0 \approx 160$ cells at one layer to about 4000 at four layers. Plotted against the diameter of the pancake at the moment of squeezing (Figure 15), the threshold varies much less: patterns appear once the pancake is about 8 to 17 cell diameters wide (roughly 80 to 170 μm). Most of the effect of the thickness is therefore that a thicker slab needs many more cells to reach the same width.

The outcome depends on when it is read. Figure 16 shows the same simulations when only 12% of the cells are B: the rings break into islands, larger aggregates polarise late, and the smallest

Layers	Polarised up to N_0	Central cluster	Patterns from N_0
1	–	75–150	≈ 160
2	75–89	106–424	≈ 500
3	300	424–1200	≈ 1300
4	300–600	424–3394	≈ 4000
6–12	849–1200	1200–2400	none up to 2400
unconfined	1200	1200–4800	none up to 4800

Table 1: The thresholds of Figure 14.

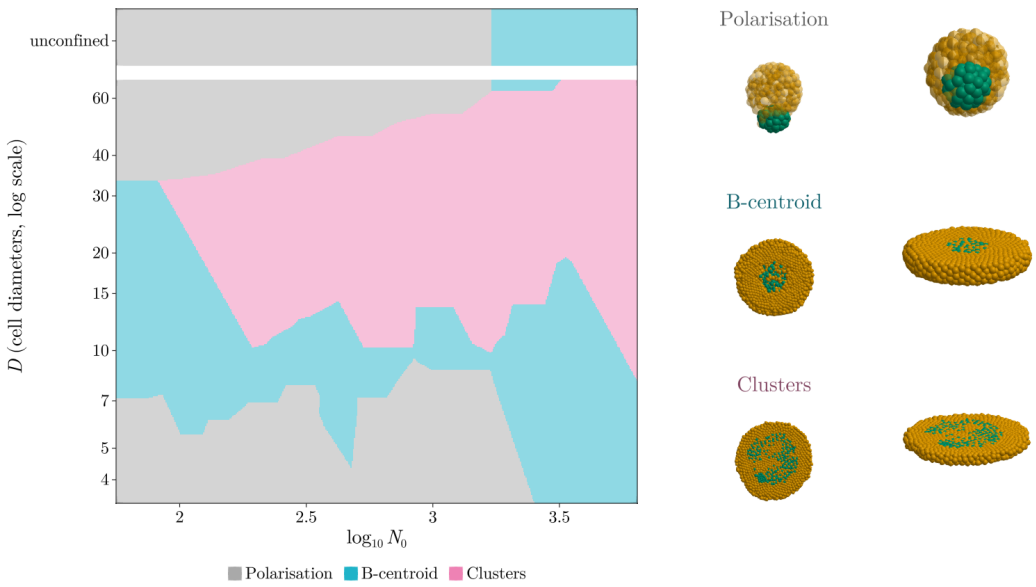


Figure 15: The same phase space against the diameter of the pancake at the moment of squeezing, in cell diameters (read when 12% of the cells are B).

monolayer patterns shrink back to a central cluster.

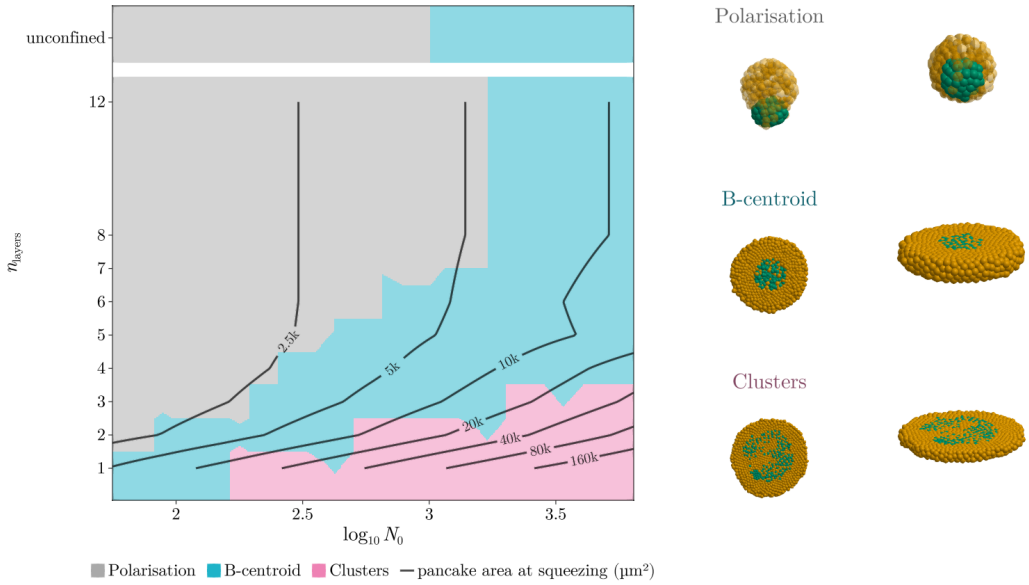


Figure 16: The same simulations, read earlier, when 20% of the cells are B (18 h after the start of differentiation on average): the readout used until November 2.

3.2 Analysis of the outcome

The outcome is read from the positions $\mathbf{x}_i$ and states s_i of the cells when 12% of them are B, about 26 h after the start of differentiation, as the experiments read at 96 h, 24 h after squeezing, with about 12% of T^+ cells. It uses the experiments' clustering and polarisation (see *The experiment*), adapted to the size of our aggregates.

Clusters. Let $\rho = 2.6\,r = 13$ µm and let $\mathcal{N}_i = \{\, j \neq i : |\mathbf{x}_i - \mathbf{x}_j| \leq \rho \,\}$ be the neighbours of cell i. A B cell is a *core* cell if at least a fraction ϕ of its neighbours are B:

$$i \in \mathcal{K} \iff s_i = B, \quad \mathcal{N}_i \neq \emptyset \quad \text{and} \quad \big|\{\, j \in \mathcal{N}_i : s_j = B \,\}\big| \;\geq\; \phi\,|\mathcal{N}_i|.$$

Two core cells are linked if $|\mathbf{x}_i - \mathbf{x}_j| \leq \rho$, and the clusters are the connected components of the graph of core cells with these links. Every other B cell with a core cell within ρ joins the cluster of its nearest core cell (a *border* cell); the remaining B cells belong to no cluster. Clusters of fewer than $n_{\min}$ cells (core and border) are discarded. If $n_1, \ldots, n_K$ are the sizes of the clusters kept and $p_k = n_k / \sum_l n_l$, the effective number of clusters is

$$N_{\text{eff}} = \frac{1}{\sum_{k=1}^{K} p_k^2},$$

which is K when the clusters have the same size and close to 1 when one of them holds most of the cells.

Polarisation. With $\bar{\mathbf{x}} = \frac{1}{N}\sum_i \mathbf{x}_i$ the centre of mass of the aggregate and $\bar{\mathbf{x}}_B$ that of the B cells, let $\mathbf{e}_1$ be the first principal axis of the cell positions (the leading eigenvector of $\frac{1}{N}\sum_i (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top$) and $u_i = (\mathbf{x}_i - \bar{\mathbf{x}}) \cdot \mathbf{e}_1$. Then

$$P = \frac{|\bar{\mathbf{x}}_B - \bar{\mathbf{x}}|}{R}, \qquad R = \tfrac{1}{2}\big(\max_i u_i - \min_i u_i\big).$$

Outcome. At a given moment, in this order:

1. *Polarisation*, if $P \geq 0.3$;
2. *Clusters*, if $N_{\text{eff}} \geq 1.5$;
3. *B-centroid* otherwise: one dominant cluster that has not moved. A ring that has not broken is a B-centroid, as in the experiments' medium confinement.

When. Let t_f be the first moment at which a fraction f of the cells are B, and

$$c(t) = \frac{\sum_k n_k}{N_B}$$

the fraction of the B cells that are in kept clusters at time t. The outcome is the one at $t_{0.12}$ if $K \geq 1$ and $c(t_{0.12}) \geq 1/2$ there; otherwise the one at $t_{0.15}$ under the same condition; otherwise the one at $t_{0.2}$. When fewer than half of the B cells are in clusters, B is differentiating away: the islands shrink below $n_{\min}$ cells one by one, and the last one left would read as a B-centroid. The rule keeps the outcome the aggregate had before, so a central cluster that differentiates away is a B-centroid and islands that differentiate away are Clusters. In all our simulations at least one cluster is kept at $t_{0.2}$, so there is no fourth outcome.

Differences with the experiments. We keep clusters of $n_{\min} = 10$ cells instead of 20, because our aggregates are 10 to 100 times smaller (with 20, monolayers of up to 300 cells have no cluster at all). A core cell needs half of its neighbours to be B ($\phi = 0.5$) instead of 40% ($\phi = 0.4$): with 40%, a ring whose pieces still touch through threads of one or two B cells is one cluster, and with 50% these threads are border cells and the pieces are separate clusters. The radius ρ and the cutoffs $P \geq 0.3$ and $N_{\text{eff}} \geq 1.5$ are those of the experiments.

Damköhler number The Damköhler number compares how fast cells differentiate with how fast they mix,

$$\text{Da} = \frac{q\,\ell^2}{D},$$

where q is the $B \to C$ rate, D the diffusion coefficient of the cells and ℓ a length. If $\text{Da} \gg 1$, a cell differentiates long before it has moved a distance ℓ, so the B pattern is set where the cells already are. If $\text{Da} \ll 1$, cells mix over ℓ before they differentiate.

There is no noise in the model: cells move only because their neighbours divide and push them, and because differentiation changes the adhesion. We therefore measure D in growing monolayers (one layer, $N_0 = 600$ and 2400, two seeds each), saving one frame every time unit (4 minutes). Each cell is followed between two of its divisions, and only its in-plane displacement is used. The spreading of the slab as it grows is not diffusion, so it is removed from every displacement with an affine fit of the displacement field, $\mathbf{u} = A\,\mathbf{r} + \mathbf{b}$. As a check we also use the displacement relative to the neighbours within $5r$. Cells in the outer 20% of the radius are left out. The mean squared displacement is then fitted with a straight line at long lag times (2 to 8 h),

$$\langle |\Delta \mathbf{r}|^2 \rangle = 4D\tau + c.$$

The MSD is nearly linear from 4 min to 10 h (Figure 17). Between 2 and 8 h its log–log slope is 1.2–1.3, so some directed motion remains: local flows that a single affine fit cannot remove, such as B growing faster than C and the sorting of the two. The sizes and the seeds agree within 20%:

$$D \approx 0.004\,r^2 \text{ per time unit} \approx 1.5\ \mu\text{m}^2/\text{h}$$

for B and C cells alike (0.0025–$0.0034\,r^2$ per time unit relative to the neighbours).

With $q = 0.004667$ per time unit:

- for ℓ one cell diameter, Da $\approx$ 4–8: a cell needs about 17 h to move by its own diameter, longer than a B cell lives ($1/q \approx 14$ h);
- for ℓ the radius of the pancake at squeezing, Da $\approx$ 500–1000 at $N_0 = 600$ and 2100–3800 at $N_0 = 2400$.

Differentiation is therefore much faster than single-cell mixing at every size. For a fixed D, Da grows with the pancake area (Da $\approx 0.015\,A/\mu\text{m}^2$ for a monolayer), so the iso-area lines of the phase space are also lines of constant Da, and the pattern threshold, at $A \approx 1.5 \times 10^4$ μm^2, sits at Da ≈ 200.

Remark 3. *Over one frame (4 min) a cell moves about* $0.04\,r \approx 0.2$ *μm, i.e.* 0.05 *μm/min. This is about 25 times slower than the instantaneous cell speed* $V_0 \approx 1.25$ *μm/min used to set the time unit* $T_0 = R_0/V_0 = 4$ *min.*

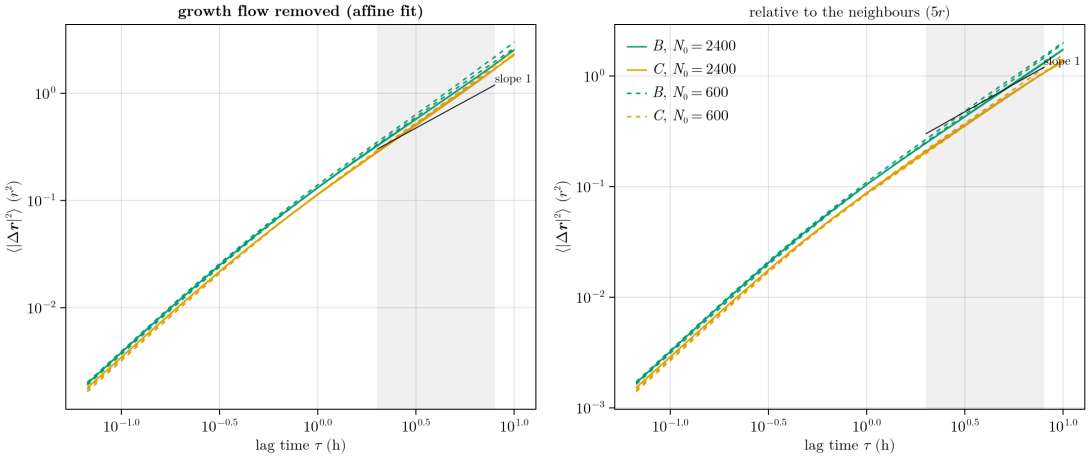


Figure 17: Mean squared displacement of B (green) and C (yellow) cells in growing monolayers ($N_0 = 600$ dashed, 2400 solid). Left: growth flow removed with an affine fit; right: relative to the neighbours within $5r$. Shaded: the lag times of the fit.

4 Open points and possible improvements

1. Large aggregates that are not confined (or confined with a wide gap) neither pattern nor polarise before B runs out. A free B core can always round up, so it does not break into islands.
2. The transition from a central cluster to a ring is gradual: near the threshold the cluster becomes loose and starts opening. Where we put the cut moves the thresholds by about one step in N_0.
3. The value $\mu_{BB} = 1.9$ is close to the point where B dissolves into C ($\mu_{BB} = 1.85$).
4. More simulations per point would sharpen the boundaries; the phase space has 2 or 3 per point, and 1 at $N_0 = 4800$.

5 Parameters

Table 2: Parameters of Oriola et al. 2026 (DO), of Sotiris' fusion experiments (Sotiris) and of this report (Current). Dark green: Sotiris differs from DO (also in Current, where Current keeps Sotiris' value). Dark blue: Current differs from Sotiris. Matrices are ordered (A, B, C) and symmetric.

Symbol	Description	DO	Sotiris	Current
Δt	Timestep	0.02	0.02	0.015
r	Radius	1	1	1
	Overlap factor	0.3	0.3	0.3
α	Cell-ECM	0.9	0.9	0.9
τ^A_{div}	Average division time	300	300	–
τ^B_{div}	Average division time	300	300	300
τ^C_{div}	Average division time	750	750	750
σ_{div}	Division time variability	0.5	0.5	0.5
λ	Global friction coefficient	1	1	1
μ	Force range factor	2 (all pairs)	$\begin{pmatrix} 2 & 1.82 & 2 \\ & 2 & 2 \\ & & 2.14 \end{pmatrix}$	$\begin{pmatrix} \cdot & \cdot & \cdot \\ & 1.9 & 2 \\ & & 2.14 \end{pmatrix}$
	Neighbour range factor	–	1	1
F_{rep}	Repulsion force	$\begin{pmatrix} 3 & 3 & 3 \\ & 3 & 3 \\ & & 5.1 \end{pmatrix}$	$\begin{pmatrix} 9 & 3 & 3 \\ & 6 & 3 \\ & & 6 \end{pmatrix}$	$\begin{pmatrix} \cdot & \cdot & \cdot \\ & 4.5 & 3 \\ & & 6 \end{pmatrix}$
F_{att}	Attraction force	$\begin{pmatrix} 6 & 6 & 2.4 \\ & 6 & 1.6 \\ & & 4.8 \end{pmatrix}$	$\begin{pmatrix} 3.6 & 4 & 2.4 \\ & 6.4 & 1.6 \\ & & 3 \end{pmatrix}$	$\begin{pmatrix} \cdot & \cdot & \cdot \\ & 6.4 & 1.6 \\ & & 3 \end{pmatrix}$
p	Fixed transition rate from A to B	0.009333	0.009333	–
q	Fixed transition rate from B to C	0.004667	0.004667	0.004667
K	Feedback parameter	0.2	0.2	–
λ_{wall}	Plate friction	–	–	1.5
n_ℓ	Number of layers			
L	Plate gap	–	–	$2.1\, r\, n_\ell$
$\phi_B(0)$	Initial proportion of B cells	–	–	0.6
T_0	Typical time scale	4 min	4 min	4 min
R_0	Typical length scale	5 µm	5 µm	5 µm
F_0	Typical force scale	100 pN	20 pN	20 pN
n_i	Number of neighbours			
n_i^X	Number of neighbours in state X			
ϕ_X	Proportion of cells in state X			

References

Oriola, David et al. (2026). "Collective fate decisions and cell rearrangements underlie gastruloid symmetry breaking". In: *bioRxiv*. DOI: 10.1101/2024.12.16.628776. eprint: https://www.biorxiv.org/content/early/2026/02/24/2024.12.16.628776.full.pdf. URL: https://www.biorxiv.org/content/early/2026/02/24/2024.12.16.628776.